

Review of final report:
International Research Training Group
DROPIT: “Droplet Interaction Technologies
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As quoted from the final report: “The research of DROPIT had been organized in three thematic areas (TA): TA-A: Drop-Gas Interaction, TA-B: Drop-Wall Interaction, TA-C: Drop-Liquid Interaction.”

The results of this study have many applications: atomization and spray in internal combustion engines, aircraft icing, sloshing in cryogenic fuel tanks, nuclear reactor cooling systems, microfluidic devices, additive manufacturing, hydraulic fracturing, spray cooling, medical inhalers, human perspiration, PEM fuel cells, membrane emulsification and filtration, thermal insulation, air-conditioning of buildings, distribution of plant protection agents, coating processes, gluing, fire suppression, medical sprays, shelf life of food, asphalt preservation, biomass, ceramic green bodies, soil erosion, encapsulation technologies, and the design of “breathable” fabrics.

I find that all of the published articles and PhD Thesis, that were supported by way of the DFG “Dropit” program, make important contributions towards furthering our understanding of the fundamental mechanisms governing multi-phase and multiphysics flows in general, and droplet technologies in particular. I highlight below some of the research here which report on studies in which recent experimental results were compared to novel simulations and mathematical modeling.

- Palmetshofer, P., Saha, R., Bryce, J., Geppert, A., Cossali, G.E., Weigand, B., 2025. Droplet impact onto sub-millimeter microstructures fully and barely submerged in a liquid film. *Phys. Fluids* 37, 082130.
- Palmetshofer, P., Wurst, J., Geppert, A., Schulte, K., Cossali, G.E., Weigand, B., 2025. Wetting behavior in the inertial phase of droplet impacts onto sub-millimeter microstructured surfaces. *Journal of Colloid and Interface Science* 682, 413422.
- Palmetshofer, P., Geppert, A., Steigerwald, J., Arcos Marz, T., Weigand, B., 2024. Thermocapillary central lamella recess during droplet impacts onto a heated wall. *Sci Rep* 14, 1102.
- Schubert, S., Steigerwald, J., Geppert, A., Weigand, B., Lamanna, G., 2024. Micro-PIV study on the influence of viscosity on the dynamics of droplet impact onto a thin film. *Exp Fluids* 65, 69.

- Geppert, A., Stober, J., Steigerwald, J., Schulte, K., Tonini, S., Lamanna, G., 2024. Analyzing the early impact dynamics of single droplets impacting onto wetted surfaces. *Phys. Fluids* 36, 012005.
- Veyskarami, M., Bringedal, C., Helmig, R., 2024. Modeling and Analysis of Droplet Evaporation at the Interface of a Coupled Free-Flow-Porous Medium System. *Transp Porous Med* 151, 27452775.
- Ackermann, S., Fest-Santini, S., Veyskarami, M., Helmig, R., Santini, M., 2023. Experimental validation of a coupling concept for drop formation and growth onto porous materials by high-resolution X-ray imaging technique. *Int J Multiphase Flow* 160, 104371.
- Lamanna, G., Geppert, A., Bernard, R., Weigand, B., 2022. Drop impact onto wetted walls: an unsteady analytical solution for modeling crown spreading. *J. Fluid Mech.* 938, A34.
- Michalkowski, C., Veyskarami, M., Bringedal, C., Helmig, R., Schleper, V., 2022. Two-phase Flow Dynamics at the Interface Between GDL and Gas Distributor Channel Using a Pore-Network Model. *Transp Porous Med* 144, 429458.
- Liu, Y., Geppert, A., Chu, X., Heine, B., Weigand, B., 2021. Simulation of an Annular Liquid Jet with a Coaxial Supersonic Gas Jet in a Medical Inhaler. *Atomization Spray* 31, 95116.
- Ren, W., Foltyn, P., Geppert, A., Weigand, B., 2021. Air entrapment and bubble formation during droplet impact onto a single cubic pillar. *Sci Rep* 11, 18018.
- Steigerwald, J., Geppert, A., Weigand, B., 2021. Numerical study of drop shape effects in binary drop film interactions for different density ratios, in: *Proceedings of 15th International Conference on Liquid Atomization and Spray Systems (ICLASS)*.
- Baggio, M., Weigand, B., 2019. Numerical simulation of a drop impact on a superhydrophobic surface with a wire. *Phys. Fluids* 31, 112107.
- Fasoulas, S., Munz, C.-D., Pfeiffer, M., Beyer, J., Binder, T., Copplestone, S., Mirza, A., Nizenkov, P., Ortwein, P., Reschke, W., 2019. Combining particle-in-cell and direct simulation Monte Carlo for the simulation of reactive plasma flows. *Phys. Fluids* 31, 072006.
- Geppert, A., Chatzianagnostou, D., Meister, C., Gomaa, H., Lamanna, G., Weigand, B., 2016. Classification of Impact Morphology and Splashing/Deposition Limit for N-Hexadecane. *Atomization Spray* 26, 9831007.

Some possible future directions could be as follows:

- There is renewed interest in “Rotating Detonation Engines”: Jose I. Guerrero, Mirko Gamba, Quantifying combustion efficiency in rotating detonation engines using MHz-rate scanned-wavelength-modulation spectroscopy, *Combustion and Flame*, Volume 285, 2026, 114744, ISSN 0010-2180, <https://doi.org/10.1016/j.combustflame.2025.114744>, <https://www.sciencedirect.com/science/article/pii/S0010218025007795>
- The *Physics of Fluids Journal* has an upcoming special topics issue entitled “Scientific Machine Learning and Physics-Informed AI for Multiscale and Multiphysics Fluid Mechanics.”